

ASTRALINK — AI-BASED VIRTUAL CAMERA TRACKING SYSTEM FOR MOBILE FS

Coarse Alignment - Point-in-Time Performance Report

T = 1.96 s

1. TEST CONFIGURATION

Scenario

synthetic_fs

Target

TGT-01

Camera

640 x 480

FOV

4.0 deg x 3.0

Update Rate

30.0 Hz

Noise: None 10.00 | Camera jitter: OFF 5.00 px/frame | Platform motion: OFF 10.00 px | Turbulence: OFF 10.00 px

2. PERFORMANCE

Processing FPS

229.0

>= 20 FPS

Average Tracking Error

2.68 px

Maximum Tracking Error

6.08 px

<= 10 px

Acquisition Time

0.03 s

<= 2 s

Re-acquisition

- s

<= 1 s

Lock Retention

100.00 %

RMSE

2.97 px

Target Loss

0.00 %

Avg Processing

2.13 ms

Re-acquisitions

0

3. PS REQUIREMENT CHECK

Parameter	Measured	Requirement	Status
Update Rate	30.0 Hz	>= 20 Hz	PASS
Processing Speed	229.0 FPS	>= 20 FPS	PASS
Acquisition Time	0.03 s	<= 2 s	PASS
Tracking Error (Max)	6.08 px	<= 10 px	PASS
Target Loss	0.00 %	< 5 %	PASS
Re-acquisition	- s	<= 1 s	-

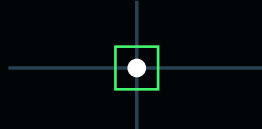
4. VISUAL SNAPSHOT (T = 1.96 s)

RADAR



Target / Boresight / Prediction

VIRTUAL CAMERA



Target TGT-01 • Error - px

3D DIGITAL TWIN



Camera terminal -> designated beacon

State: BENCHMARK | Pan: -0.17 deg | Tilt: -0.17 deg | Al: -

5. SYSTEM SUMMARY

COARSE ALIGNMENT CONTINUES - HANDOFF NOT READY

Designated target TGT-01 at 1.96 s. Detection, AI verification, Kalman prediction and pan/tilt control are reported from the recorded telemetry snapshot.